

Gastric Cancer Dx and Staging

I'm Dr Jonathan Salo, I'm a GI Cancer Surgeon in Charlotte, North Carolina. These videos are designed to educate you about cancer and its treatment and help you and your cancer care team make the right decisions for you. . If you or someone close to you has recently been diagnosed with gastric cancer, this video is intended for you.

To start, gastric cancer is cancer of the stomach. The terms can be used interchangeably

Food passes from the throat to the esophagus into the stomach. After leaving the stomach, it passes into the duodenum and finally into the jejunum, the first part of the small intestines.

The focus in this video is on cancers of the stomach.

As it turns out, cancers can also form in the esophagus.

To make things more complicated, cancers can also form in the area where the esophagus and stomach meet, called the gastroesophageal junction.

The cancers of the gastroesophageal junction behave similar to cancers of the esophagus, so they are often grouped together.

We have other videos that focus on cancer of the esophagus and gastroesophageal junction

This video focuses on cancers of the stomach, or gastric cancers.

Let's review some cancer terms. A **tumor** is an abnormal growth.

Tumors can be either benign or malignant

A **benign** tumor of the stomach may grow over time, but it won't ever spread anywhere else

A **malignant** tumor of the stomach has the *potential* to spread elsewhere in the body. This could be to the lymph nodes, the lungs, the liver, so the bone *or* spread inside the abdominal cavity..

Cancer is another term for a malignant tumor

0.1 EGD

In most cases, the diagnosis of esophageal cancer is made by endoscopy, also called an EGD for esophago-gastroduodenoscopy

Under sedation, a flexible scope is passed through the mouth into the stomach, which allows viewing the inside of the stomach.

If a tumor is found in the stomach, a small portion can be removed, which is called a **biopsy**

The biopsy is examined by a pathologist, who will determine whether the tumor is benign or whether it is a cancer

If the biopsy shows cancer, the next step is staging. Staging is the process of finding out the *size* of the tumor and whether or not there has been spread to the lymph nodes or other places in the body.

The cancer treatment is determined by the stage

Once the stage has been determined, it will be possible to determine the best therapy

Staging is going to focus on three areas:

The first is the *tumor*. How large is the tumor? And more importantly, how deeply has it grown into the wall of the esophagus?

The second is whether there is spread to the *lymph nodes*.

The third is whether there is spread to other parts of the body

spread to other parts of the body is called *metastasis*

We will start by looking at the wall of the stomach. The wall of the stomach has multiple layers, shown here.

(pause for “video”)

The inner layer is the mucosa, (32)

followed by the submucosa,

followed by the muscularis

Outside of the muscularis is the serosa.

Surrounding the stomach are lymph nodes. The purpose of lymph nodes is to filter the blood and help fight infections, but in some cases cancers from the stomach can spread to the lymph nodes

In its earliest stages, cancer of the stomach start in the inner, or most superficial layer, called with mucosa. (33)

With time, however, the cancer can continue to grow and invade deeper into the wall of the esophagus. (34)

(pause) - (35)

(pause (36)

The deeper the cancer invades into the wall of the esophagus, the more likely it is that cancer cells can spread to lymph nodes (37)

To review, the stage consists of 3 parts: (38)

- T for Tumor
- N for Nodes
- M for Metastasis or spread to other organs

We will first consider the tumor (39)

T1a tumors invade just the top layer, or mucosa (40)

T1b tumors invade into the submucosa (41)

T2 tumors invade into the muscularis (42)

T3 tumors go all the way through the muscularis (43)

It can be difficult to tell how deep a tumor has invaded just by looking from the surface.

For small tumors, *endoscopic ultrasound* can be helpful. This is an endoscopic procedure done under sedation much like a EGD and it can help determine the T stage of a stomach tumor. (44)

The EGD scope has an ultrasound probe on the end, which can be used to examine the tumor (45)

Evaluation of the *lymph nodes* is the second part of staging (45)

to determine how thick the tumor is and assign it a T stage.

- T1 tumor involves the top layers of the esophagus
 - T2 tumor invades into the muscular layer
 - T3 tumor invades all the way through the muscular layer
 - T4 tumor invades into nearby structures such as the aorta or the airway
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This may sound confusing, but as a general rule, if someone with esophageal cancer has difficulty swallowing, the tumor is usually a T3.

There are of course exceptions, but this is a general guideline

0.2 Nodes

- N0 – No lymph nodes involved
 - N+ - Lymph nodes involved with cancer
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1 Metastasis

The M classification refers to metastasis Metastasis = spread of cancer to other organs such as the lungs, liver, or bone

- M0 – No signs of spread to other organs
 - M1 – Spread to other organs
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2 Diagnostic Tests

Next we will talk about some of the diagnostic tests that are used for staging.

When you meet with your doctors, one the first things they will do is to come up with a plan for testing that is tailored to you and your particular tumor. So we will talk about these tests in general terms, but not all tests are needed for all patients

3 Scans

A CT scan is usually the first test for staging. This will show whether there is any signs of metastasis or spread to other organs such as the lung, liver, or bone

4 PET Scan

A PET scan is a specialized scan which combines a CT scan with an injection of a small amount of tracer which lights up areas of cancer.

5 EUS

In cases where it is important to know about the exact size of the tumor, an endoscopic ultrasound exam can be done. This procedure is similar to an EGD, but the endoscope has an ultrasound sensor on the end of the scope which produces an image of the tumor

In some cases, particularly for cancers in the stomach, it is important to look for signs of spread in the abdominal cavity. In some situations, cancers can spread in the abdominal cavity but the areas are so small they don't show up on a CT scan. In these cases, a laparoscopy is helpful.

5.1 Laparoscopy

Laparoscopy is a surgical procedure done under a general anesthetic. Several incisions ¼” long are made, and a telescope is inserted into the abdominal cavity. This allows an examination of the abdominal cavity to look for signs of spread.

The procedure is usually done as an outpatient, so you can go home the same day.

The next video in our series discusses your esophageal cancer care team:

[Esophageal Cancer Care Team](#)

We hope you have found this video helpful. This videos and others like it are designed to educate patients and families about esophageal cancer

and equip them for their discussions with their esophageal cancer care team.

As always, these videos are no substitute for expert medical advice.

Feel free to leave a comment or a question, or if you have suggestions for future videos.
